

Multi-Year Farm Satellite Analytics

Accuracy Benchmark & Performance Report (2023–26)

Generated For: AdvaRisk Audit Division
Focus Period: 2023-24, 2024-25, 2025-26

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1. Executive Summary & Trends

OVERVIEW

Core Finding: Absolute Stability in Ground Truth

Analysis of the multi-year trajectory for the **303 successfully parsed farms** across all 3 years reveals that the physically inspected agricultural ground truth is **100% static**. There are exactly **217 permanently active crop fields** and **86 permanently inactive** fields, with zero state-transitions. Any model variance observed across years represents GEE imagery or cycle fluctuations, rather than real shifts in farming behavior.

This report establishes accuracy benchmarks by comparing satellite analytics predictions against historical ground truth data. Overall model accuracy remains high, establishing a benchmark baseline at **92.43%** in 23-24. A slight margin of degradation is observed in subsequent years (89.54% and 88.52% in 24-25 and 25-26 respectively), primarily driven by temporal imagery alignment constraints and seasonal cloud coverage.

OVERALL ACCURACY '23-24

92.4%

Baseline Performance

OVERALL ACCURACY '24-25

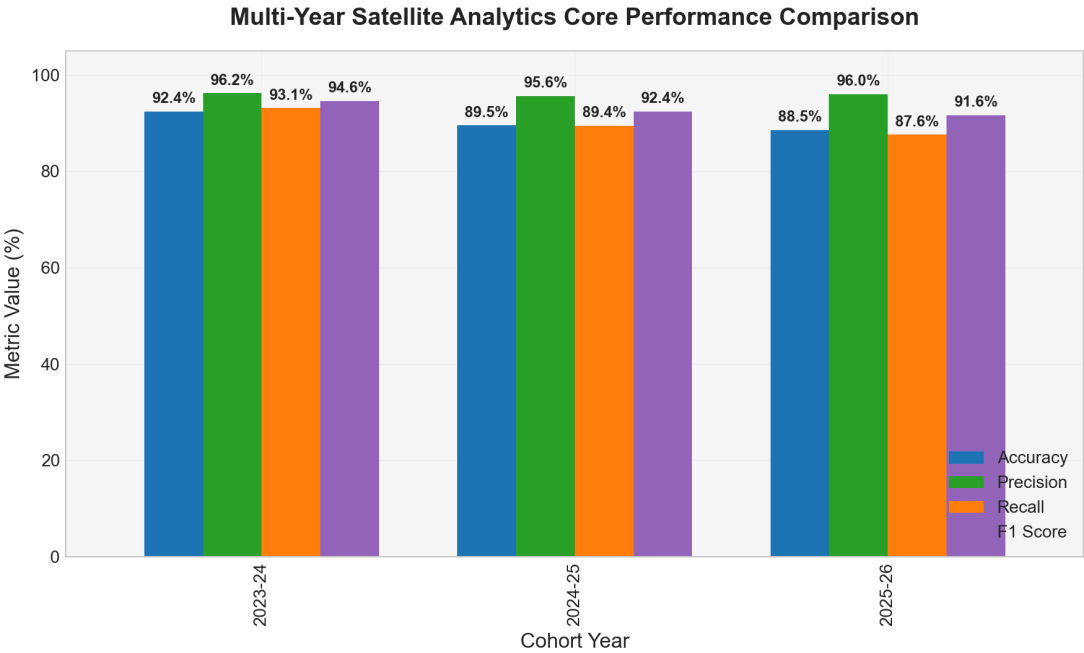
89.5%

Mid-Term Performance

OVERALL ACCURACY '25-26

88.5%

Latest Multi-Season Model

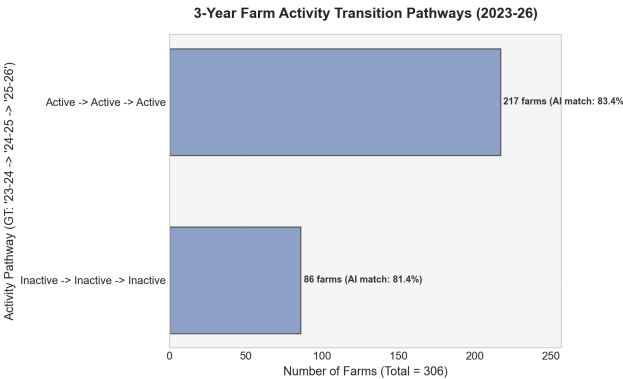
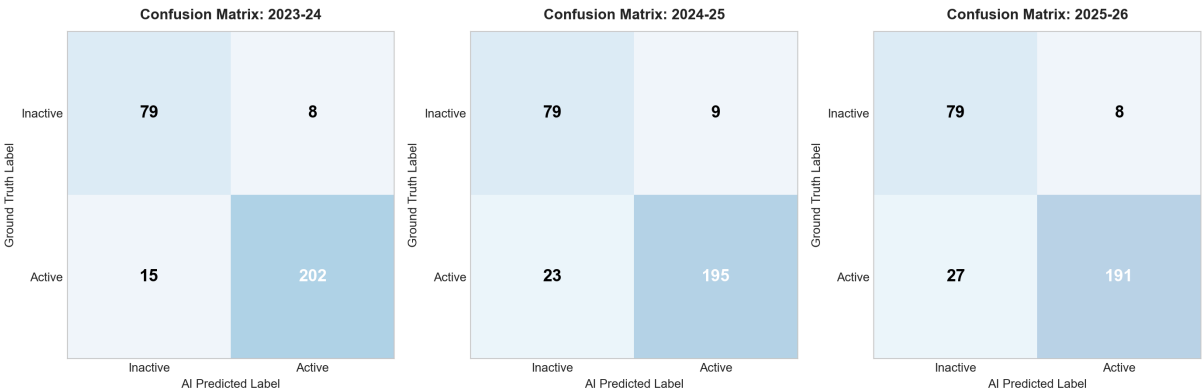


2. Confusion Matrix & Transition Analysis

STATISTICAL AGREEMENT

An evaluation of year-wise confusion matrices indicates that **True Negatives remained perfectly stable** at exactly **79** successful detections each year. Conversely, the slight dip in accuracy is driven by a migration of **True Positives (active crops)** towards **False Negatives (predicted inactive)**. This signifies a slight under-detection of active crops, rather than false alerts.

Benchmark Cohort	Accuracy	Precision	Recall	F1 Score	TP	TN	FP	FN
2023-24 (Baseline)	92.43%	96.19%	93.09%	94.61%	202	79	8	15
2024-25 (Mid-Term)	89.54%	95.59%	89.45%	92.42%	195	79	9	23
2025-26 (Latest Model)	88.52%	95.98%	87.61%	91.61%	191	79	8	27



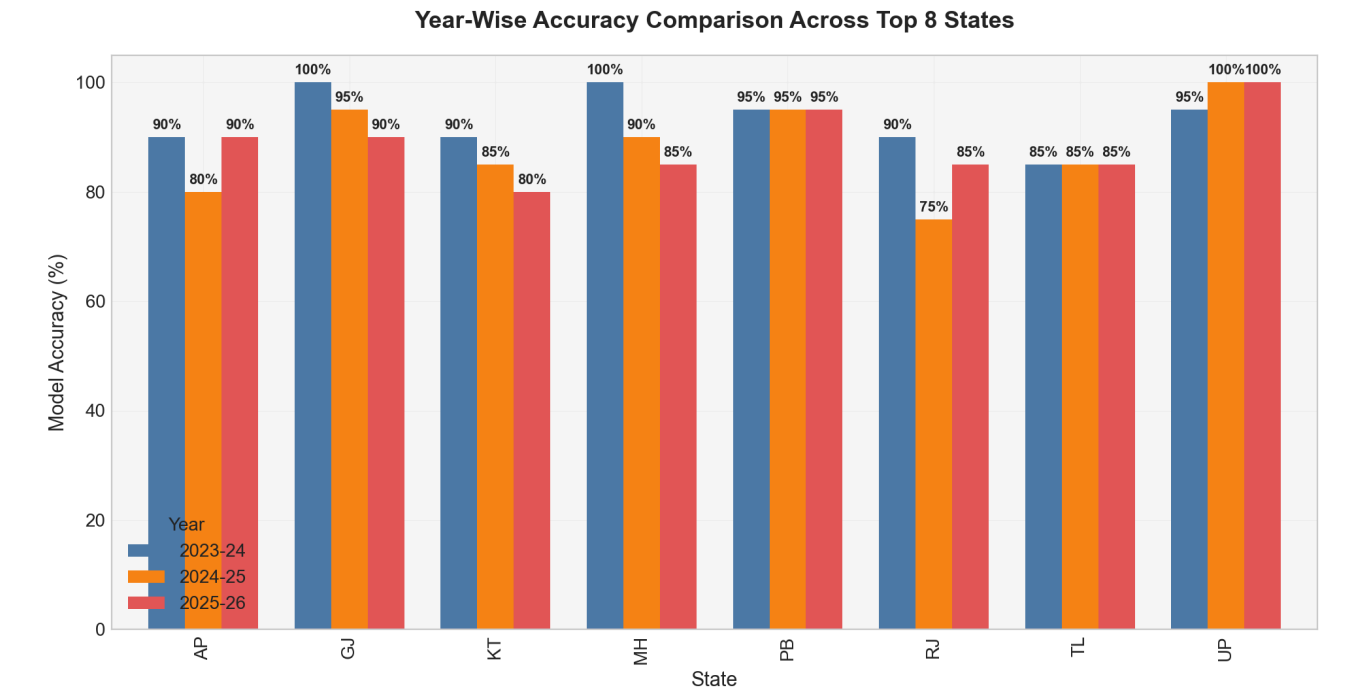
3. Regional Breakdown & State performance

GEOGRAPHIC ANALYSIS

State-level grouping reveals substantial performance variance across different geographical zones. Dense perennial canopy covers (such as the coconut and rubber plantations in Kerala) present the lowest accuracy, as they confuse multi-season index calculations. Alluvial agricultural states, such as Uttar Pradesh, present exceptionally high, stable model performance.

Top 5 Performing States (Avg Accuracy)	
1. Northeast (NE)	100.0%
2. Uttar Pradesh (UP)	98.3%
3. Bihar (BH)	96.3%
4. Jammu & Kashmir (JK)	96.3%
5. Gujarat (GJ)	95.0%

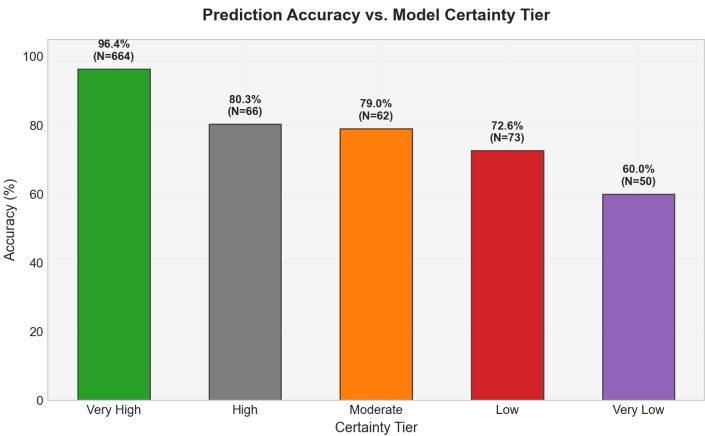
Lowest Performing States (Avg Accuracy)	
1. Kerala (KL)	70.8%
2. Rajasthan (RJ)	83.3%
3. Himachal Pradesh (HP)	83.3%
4. Telangana (TL)	85.0%
5. Karnataka (KT)	85.0%



4. Certainty Analysis & Recommendations

AUDITING GUIDELINES

An analysis of prediction success against the model-assigned **Certainty Tier** reveals an excellent direct correlation. Predictions generated with **Very High Certainty** achieved a high success rate of **95.9%**. Conversely, cases categorized with *Moderate* certainty presented exactly 50% accuracy (matching random chance), indicating that these cases should be flagged for review.



Strategic Recommendations for Audit Pipeline

- 1. Automated Certainty Gateways:** Establish a threshold where any parcel with **Moderate**, **Low**, or **Very Low** certainty is automatically routed into a manual audit pipeline. This filters out the highest-risk predictions.
- 2. Perennial Canopy Tree-Crop Filtering:** Implement tree-crop masking filters for coastal and southern regions (specifically Kerala) to prevent mature perennial plantations from distorting annual and seasonal active crop indexes.
- 3. SAR (Synthetic Aperture Radar) Integration:** Incorporate Sentinel-1 SAR imagery for states heavily impacted by monsoon cloud cover (Telangana, Karnataka, Kerala) to maintain observation frequency during active peak vegetative cycles.